Recommendations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Teach recommendation as an explained, bounded process.

Model filtering and ranking separately, audit exposure feedback and reserve the new activity task for independent evidence.

Keep one habit: Look for verified requirements, honest tradeoffs and feedback within its recorded context.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Apply **eligibility** requirements before **ranking** candidates by **preference**s

This is an explicit paper recommender, not a claim that all recommenders use the same algorithm. Separate **eligibility**, scoring, **exposure** and feedback meaning.

The feedback-loop concept is supported by the primary research Degenerate **Feedback Loop**s in Recommender Systems, https://arxiv.org/abs/1902.10730. All task records are fictional.

By the end, I can filter requirements before **ranking preference**s and avoid overinterpreting clicks.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Book records J through M and requirement and **preference** cards
- A feedback log with item, **exposure**, event and optional stated reason
- Activity records A through D for the independent task

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate similarity using explicitly selected features and a stated scoring rule

Reference example, not the only acceptable wording: A: 1, 1, 0 gives 2. B: 0, 0, 1 gives 1. C: 1, 0, 0 gives 1. A is the only highest scoring candidate among A, B and C under equal weights.

Without opening the old lesson, explain one key idea from **Prediction** Is Not Fact. Give an example and a mistake that the idea helps you avoid.

Look for: Separate a recorded **prediction** from the **observation** used to check it

Reference example, not the only acceptable wording: **Input** height 6 cm; **prediction** far because 6 is at least 6; observed distance 3 units; observed category near because 3 is below 4; mismatch.

Height is available to the model before the trial, while travel distance is measured afterward to check its **prediction**.

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

Look for: Construct a classifier with a stated **input** format, selected rule and recorded **output**

Reference example, not the only acceptable wording: Require a present height, unit cm, a whole number and range 1 through 10 inclusive.

For example, 7 cm is accepted; blank is missing, 60 mm has the wrong unit and 11 cm is outside the range. Fractional values such as 4.5 cm also fail.

The contract defines accepted **input**s; it does not list only the **fitting** records or establish **prediction** correctness.

No. **Eligibility** requirements must be checked separately. A high **preference** score does not make an ineligible item suitable.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Similarity

Prediction Is Not Fact

Build a Human Classifier

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A system shows only adventure books and records only adventure clicks. It concludes the reader dislikes poetry. Explain the missing **evidence** and propose a small test.

Concept to explain

A recommender ranks possibilities using goals, information, and **tradeoff**s.

Term	Meaning
ranking	ordered list from higher to lower priority
objective	goal the scoring process tries to satisfy
tradeoff	gain in one goal that may reduce another
diversity	meaningful variety among recommended options

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach requirements, ranking and feedback as separate checks

> Use the exact rule to explain both exclusions and remaining preference tradeoffs.

Begin with BookRec1's complete task

A **recommendation** suggests an option for a particular purpose. Start with requirements: conditions every suggested item must meet. Then use **preference**s to rank the eligible options.

An item that fails a requirement cannot make up for it by collecting more **preference** points.

A fictional reader needs an available English book. The reader prefers topic space and format comic. BookRec1 gives eligible books 2 points for space and 1 for comic; a known mismatch gives 0.

It returns every eligible book tied for the highest score. If none are eligible, it reports no eligible option rather than ignoring a requirement.

The candidates are J: space, prose, English, available; K: animals, comic, English, available; L: space, comic, Spanish, available; M: space, comic, English, unavailable. These are invented records.

The purpose is to trace a **recommendation**, not to infer anything about real students or collect their personal **preference**s.

Trace the eligible scores and faithful explanation

J passes both requirements and scores 2. K passes and scores 1. L fails the English requirement; M fails availability. Do not place L or M above J because they would earn 3 **preference** points.

They are excluded before the eligible scores are compared.

A faithful explanation says J is available in English and has the highest eligible score because its space topic matches the stated **preference**. It should also mention that J is prose, not comic.

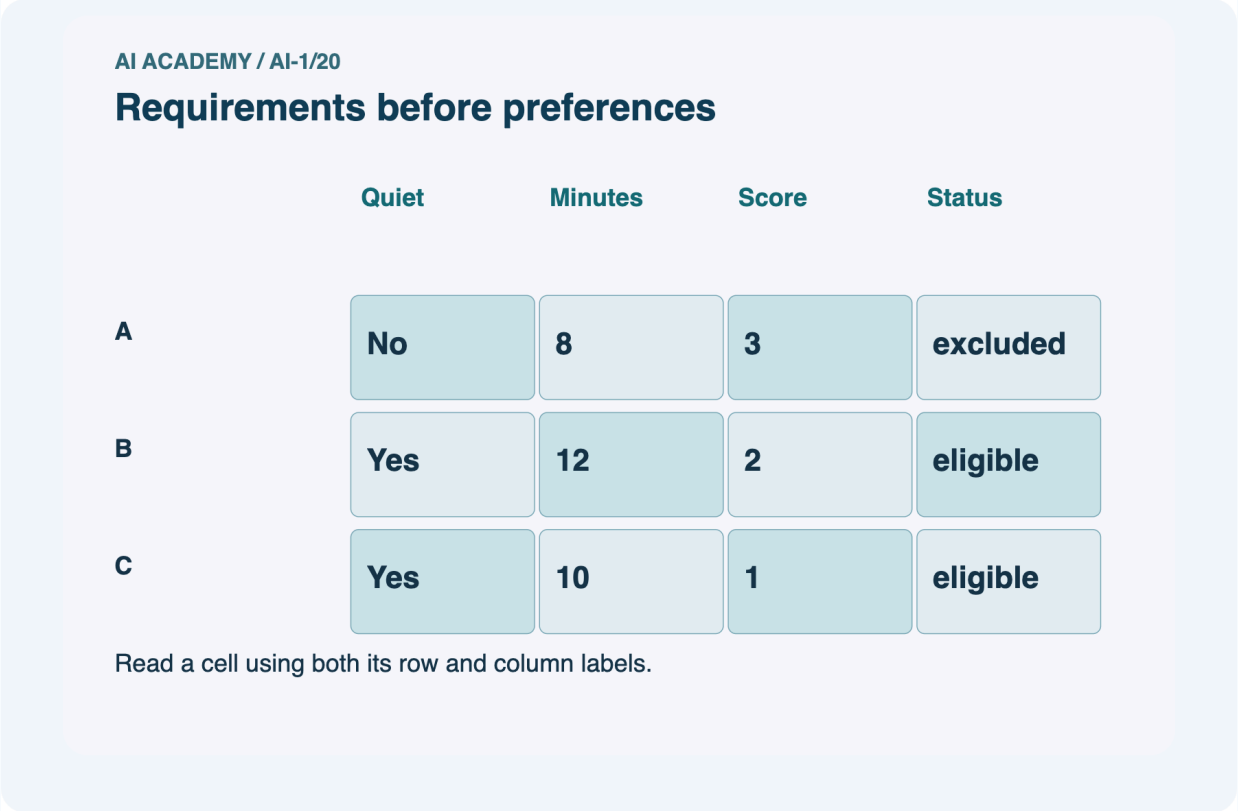
Claiming that the system knows the reader will love J would go beyond the computation and the **evidence**.

Model the separate break chooser

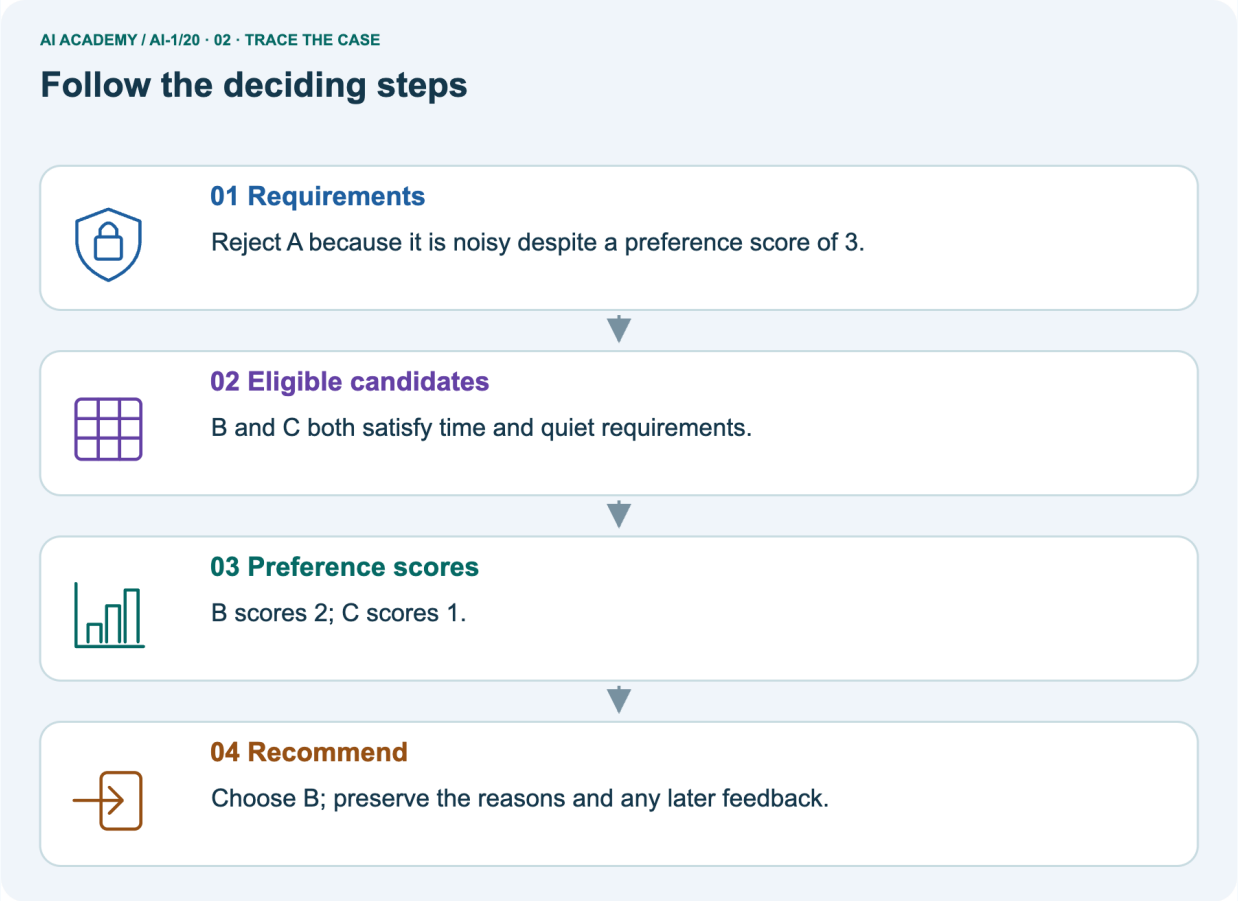
A fictional classroom break chooser recommends a suitable paper activity.

Must be quiet and take at most **12 minutes**. Score 2 for puzzles and 1 for solo.

A puzzle,solo,noisy,8 min; B puzzle,group,quiet,12 min; C drawing,solo,quiet,10 min.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

A hard requirement determines whether an option can be chosen at all. **Preference**s rank the eligible options. The apparent best **preference** match, A, is not suitable under the quiet requirement.

If a learner clicks C instead, the event alone does not tell us whether they disliked B, misclicked or wanted variety.

Change only the maximum time

1. Change: Change the maximum time to 10 minutes.
2. **Prediction**: Only C remains eligible and is recommended.
3. Reason: B takes **12 minutes**; A is still noisy.

Limit the satisfaction claim from a click

One click is an observed event, not proof of satisfaction or a stable personal **preference**.

Audit what exposure and recorded events establish

Recorded event	What it establishes	What it does not establish
J was shown	There was a recorded exposure	The reader noticed or liked it
J was clicked	A recorded interaction occurred	The reader enjoyed the book
Reader said J was too long	A stated reason in this context	All space books are unwanted
K was never shown	No exposure recorded for K	The reader rejected K

A click can reflect interest, curiosity or a mistaken tap. No click can reflect a missed display, lack of time or a decision not to choose it. Preserve the observed event separately from any interpretation.

More direct feedback may clarify a reason, but one response still describes one person and occasion.

Useful feedback records name the item, what was shown, the system version, the event and any stated reason. Use only information needed for the task.

Our paper exercises use invented feedback; they do not require accounts or recording classmates' browsing behaviour.

Use the picture deliberately

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT reader goals and candidate-book traits

2. Process — the deciding step

PROCESS score and rank candidates with chosen **tradeoff**s

3. Output — the result

OUTPUT ordered **recommendation**s plus reasons

CORE IDEA A recommender ranks possibilities using goals, information, and **tradeoff**s.

UNDER THE HOOD **Recommendation** is **ranking**, not merely classification. An **objective** combines relevance signals and constraints; **tradeoff**s may include familiarity, **diversity**, novelty, safety, and availability.

The top score reflects the **objective**, not universal quality.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

CASE Two readers can receive different book **ranking**s because their goals, reading level, interests, and novelty **preference**s differ.

STEP 1 — OBSERVE reader goals and candidate-book traits

STEP 2 — TRACE score and rank candidates with chosen **tradeoff**s

STEP 3 — CONCLUDE ordered **recommendation**s plus reasons

MODEL REASONING Use the same transparent scoring method with each reader's stated **preference**s. Different **input**s create different totals. Report **tradeoff**s, ties, and one option outside the top rank to preserve choice.

DO NOT OVERCLAIM A **recommendation** optimizes chosen goals and data; it does not discover what a person truly wants.

What new **evidence** could change this answer?

Concrete teaching case

A recommender suggests adventure books after one adventure-book choice.

Explain its **evidence** and limitation.

Exact explanation to model

The past choice supports considering adventure, but does not capture every interest. Ask for feedback and offer variety.

Explain the added worked example

Key terms

Term	Meaning
Eligibility	Whether an item meets the requirements for consideration
Preference	A desired feature used to rank eligible choices
Recommendation	A suggested option produced under a stated process
Exposure	An opportunity for a person to see an item
Feedback loop	A process in which earlier recommendations influence the feedback used for later recommendations

Model the reasoning aloud

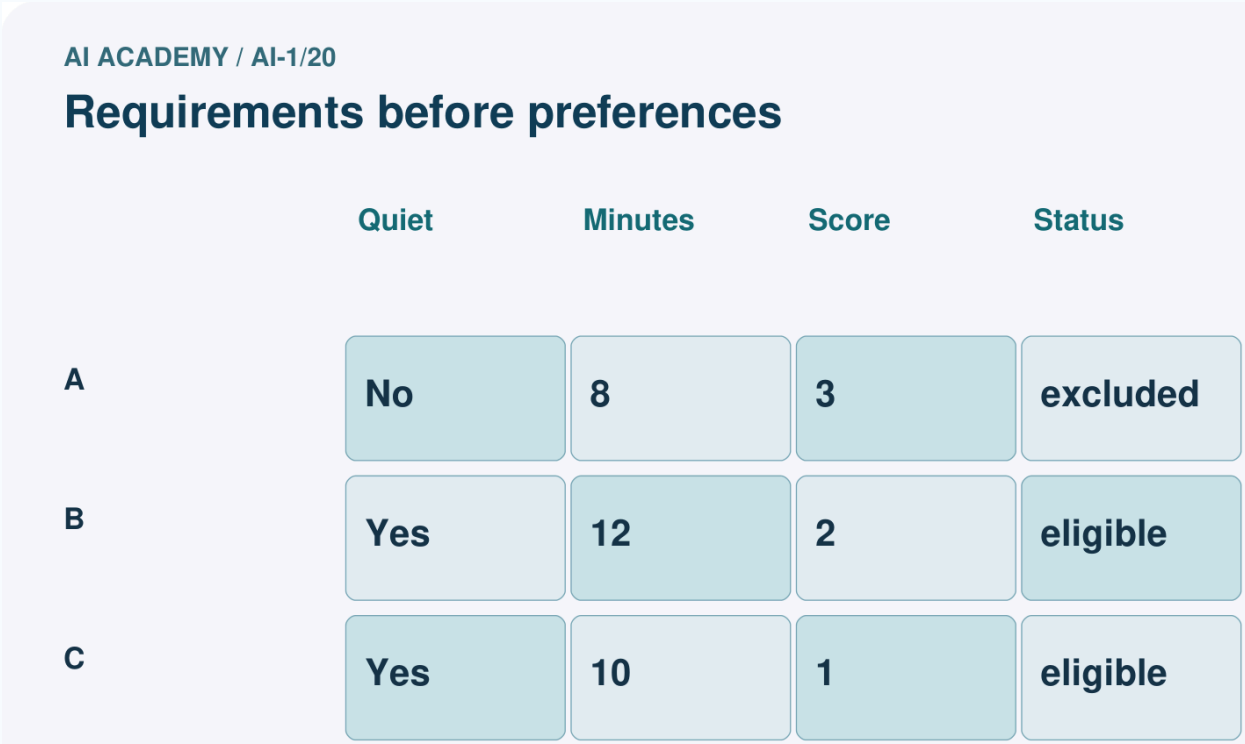
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Recommendations

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/20

Recommendations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Requirements

Reject A because it is noisy despite a preference score of 3.



2. Eligible candidates

B and C both satisfy time and quiet requirements.



3. Preference scores

B scores 2; C scores 1.



4. Recommend

Choose B; preserve the reasons and any later feedback.

CHANGE ONE THING

Change the maximum time to 10 minutes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Requirements	Reject A because it is noisy despite a preference score of 3.
2. Eligible candidates	B and C both satisfy time and quiet requirements.
3. Preference scores	B scores 2; C scores 1.
4. Recommend	Choose B; preserve the reasons and any later feedback.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Write an explanation of why J was recommended that mentions one **preference** it does not match. A marketer adds guaranteed enjoyment. Explain why that phrase is unsupported.

Expected reasoning and fading help

Teacher check: J is available in English and its space topic earns 2, the highest score among eligible choices; it is prose rather than the preferred comic. The rule supplies a **ranking**, not **evidence** guaranteeing enjoyment.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Recommendation**s and feedback. Proposed explanation: “A high score can cancel a failed requirement”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Filter on requirements before **ranking preference**s.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Why does the **recommendation** change without changing anyone's **preference**s?

The requirement changes **eligibility**. C becomes the only eligible choice even though B would still score more **preference** points.

Changed-case reasoning: Only C remains eligible and is recommended. B takes **12 minutes**; A is still noisy.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: A recommender suggests adventure books after one adventure-book choice.

Write the question to explain: Explain its **evidence** and limitation.

Reveal after **prediction**s: The past choice supports considering adventure, but does not capture every interest. Ask for feedback and offer variety.

Use the earlier changed question as optional later practice. Reserve Transfer the process to activities for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

J: shown and clicked; enjoyment not established. K: never shown; dislike not established. Ask an optional neutral question about suitability and reason, or offer eligible alternatives and record **exposure** and response.

Exit: only showing prior clicked items restricts opportunities for other feedback; include eligible alternatives and preserve which options were actually shown.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve why a similarity score cannot override a requirement.
7 to 20 minutes	Trace BookRec1 eligibility and scoring. Ask why L and M never enter the eligible ranking.
20 to 33 minutes	Complete W1 and W2. Compare a faithful explanation with a claim of guaranteed enjoyment.
33 to 45 minutes	Read the feedback table and complete W3. Preserve what was observed without guessing motives.
Session 2 start 0 to 8 minutes	Retrieve the difference between an unseen item and a rejected item.
8 to 23 minutes	Complete the guided feedback record and trace the space-only display loop. Discuss an optional, neutral feedback question.
23 to 38 minutes	Collect W4 independently. Require the tie, the 20-minute boundary and limits on the selection log.

Stage	Teacher action
38 to 45 minutes	Discuss W5 and the exit. Test no eligible option and unknown availability without relaxing requirements silently.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

For both fictional readers, only U and V are eligible; W fails language and X availability. R1 gives U=2,V=1 and recommends U. S1 gives U=1,V=2 and recommends V. If U and V availability become unknown, both are pending and no verified eligible option remains. R1's shown-and-clicked U is an exposure and interaction record, not proof of enjoyment; unshown V is not a recorded rejection. Reserve Transfer the process to activities for the individual first attempt.

Run and record

1. Complete the task: Score books for two readers; explain why **ranking**s differ.
2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

A high score can cancel a failed requirement: Filter on requirements before **ranking preference**s.

A click proves enjoyment: It is an interaction; motive and satisfaction need further **evidence**.

No clicks means dislike even without **exposure**: An unseen option has not received a meaningful opportunity for a response.

Choose support from the response

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Predict first, test honestly, and explain what the **evidence** means.

State your **prediction** before the result is known.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Transfer the process to activities for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

CLAIM TO REPAIR A **recommendation** decides what a person must choose.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

No eligible result means the system must choose anyway: Return no verified eligible option and explain exclusions or missing facts.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: A fictional learner needs an indoor activity lasting at most 20 minutes. **Preference**s score 2 for drawing and 1 for solo.

Candidates: A drawing, solo, outdoor, 10 minutes; B drawing, group, indoor, 20 minutes; C music, solo, indoor, 15 minutes; D drawing, group, indoor, 18 minutes. Filter, score and return all highest scorers.

Only B is then shown and selected; C and D are not shown. State what the feedback establishes and what remains unknown.

A excluded because outdoor. B eligible scores 2; C eligible scores 1; D eligible scores 2. Return B and D. Exactly 20 minutes passes at most 20. The log establishes B shown and selected, not enjoyment.

Preferences about unshown C and D are unknown; selecting B does not prove rejection of D or validate the choice to hide it.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Filter before scoring

Apply BookRec1 to J, K, L and M from the lesson. State **eligibility** or exclusion reason for each, the scores of eligible books and the **recommendation**. Explain why L's two **preference** matches do not make it the winner.

J eligible scores 2; K eligible scores 1; L excluded for Spanish; M excluded because unavailable. Recommend J. **Preference** matches cannot compensate for failing the English requirement.

W2 Explain a recommendation honestly

Write an explanation of why J was recommended that mentions one **preference** it does not match. A marketer adds guaranteed enjoyment. Explain why that phrase is unsupported.

J is available in English and its space topic earns 2, the highest score among eligible choices; it is prose rather than the preferred comic. The rule supplies a **ranking**, not **evidence** guaranteeing enjoyment.

W3 Do not label an unseen item rejected

J was shown and clicked. K was never shown. Separate the observed records from the labels enjoyed and disliked. Write a question that could gather a stated reason while allowing no answer.

Record J shown and clicked; K not shown. Enjoyment of J and dislike of K remain unsupported.

An optional question such as Did this suggestion suit what you wanted, and why or why not? seeks a reason without presuming satisfaction. No answer remains missing feedback.

W5 Handle an empty eligible set

Return to BookRec1. J and K become unavailable, L remains Spanish, and M's availability becomes unknown. A designer proposes recommending M anyway because it matches both **preference**s.

Apply the requirements and propose an honest response.

J and K are excluded as unavailable, L for language. M is pending availability verification and cannot yet count as eligible.

Report no verified eligible option, identify the missing availability fact, and request confirmation or let the reader revise requirements explicitly.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

The past choice supports considering adventure, but does not capture every interest. Ask for feedback and offer variety.

Additional transfer practice question

A recommender suggests a space book because you liked two space books. Does the suggestion prove you will enjoy it? What feedback could help next time?

Reasoned practice answer

No; shared topic supports a suggestion, not a guarantee. Feedback about what you liked or disliked can guide later **recommendation**s, while other interests should remain possible.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A **recommendation** decides what a person must choose.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

The user never saw poetry, so clicks cannot establish rejection of it. Offer a balanced small choice including poetry, record what was shown as well as chosen, and ask for feedback.

One choice remains limited **evidence** of a person's interests.

Explanations for the additional workbook examples

W1 Filter before scoring

W2 Explain a recommendation honestly

W3 Do not label an unseen item rejected

W4 Transfer the process to activities

A fictional learner needs an indoor activity lasting at most 20 minutes. **Preference**s score 2 for drawing and 1 for solo.

Candidates: A drawing, solo, outdoor, 10 minutes; B drawing, group, indoor, 20 minutes; C music, solo, indoor, 15 minutes; D drawing, group, indoor, 18 minutes. Filter, score and return all highest scorers.

Only B is then shown and selected; C and D are not shown. State what the feedback establishes and what remains unknown.

A excluded because outdoor. B eligible scores 2; C eligible scores 1; D eligible scores 2. Return B and D. Exactly 20 minutes passes at most 20. The log establishes B shown and selected, not enjoyment.

Preferences about unshown C and D are unknown; selecting B does not prove rejection of D or validate the choice to hide it.

W5 Handle an empty eligible set

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

If the sorter ranks bins, state the **objective** and show why the top suggestion is not a command.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Separate sorter requirements, **preference**s and observed feedback; document why a suggestion was eligible.

Support while preserving the reasoning goal

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Use separate trays for excluded, pending and eligible items. Score only the eligible tray.

Extension: change the display order while keeping scores fixed and explain why different clicks would not automatically show changed **preference**s.

W4 requires excluding A, scores B 2 C 1 D 2, both B and D returned, and no inferred enjoyment or rejection from unshown items. After support change the time limit to at most 18: B becomes excluded, D wins with 2 over C's 1.

Keep feedback limits unchanged.

Plan the delayed check

Suggested return points in this level: ai-1/21, ai-1/23, ai-1/26. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 21: Good Data, Bad Data. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 21 investigates data quality and how to repair a record with **evidence** while preserving its history.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Helpful Classroom Sorter

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Apply **eligibility** requirements before **ranking** candidates by **preference**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Write an explanation of why J was recommended that mentions one **preference** it does not match. A marketer adds guaranteed enjoyment. Explain why that phrase is unsupported.

Reasoned practice answer: J is available in English and its space topic earns 2, the highest score among eligible choices; it is prose rather than the preferred comic.

The rule supplies a **ranking**, not **evidence** guaranteeing enjoyment.

Check every learner before discussion: Use the worked case for **Recommendation**s and feedback. Proposed explanation: “A high score can cancel a failed requirement”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Filter on requirements before **ranking preference**s.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change the maximum time to 10 minutes. Require a **prediction** and a reason before comparing. Expected result: Only C remains eligible and is recommended. Reason: B takes **12 minutes**; A is still noisy.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

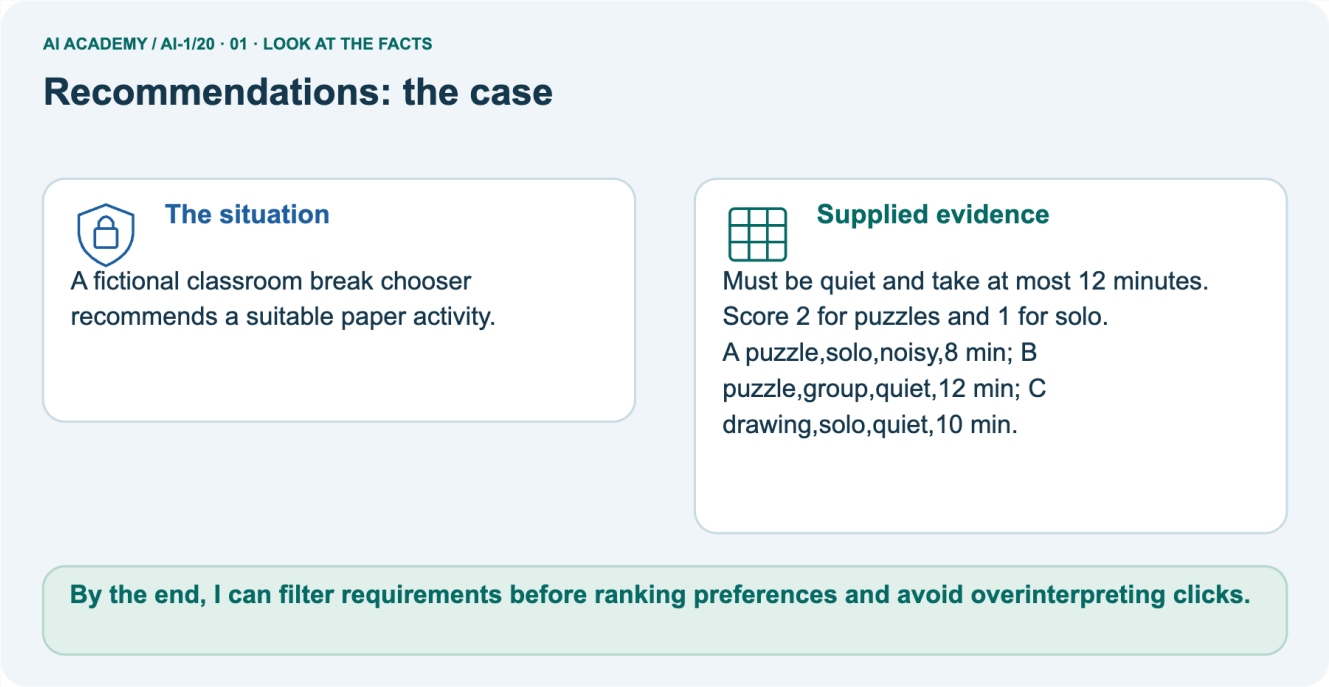
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Recommendations: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

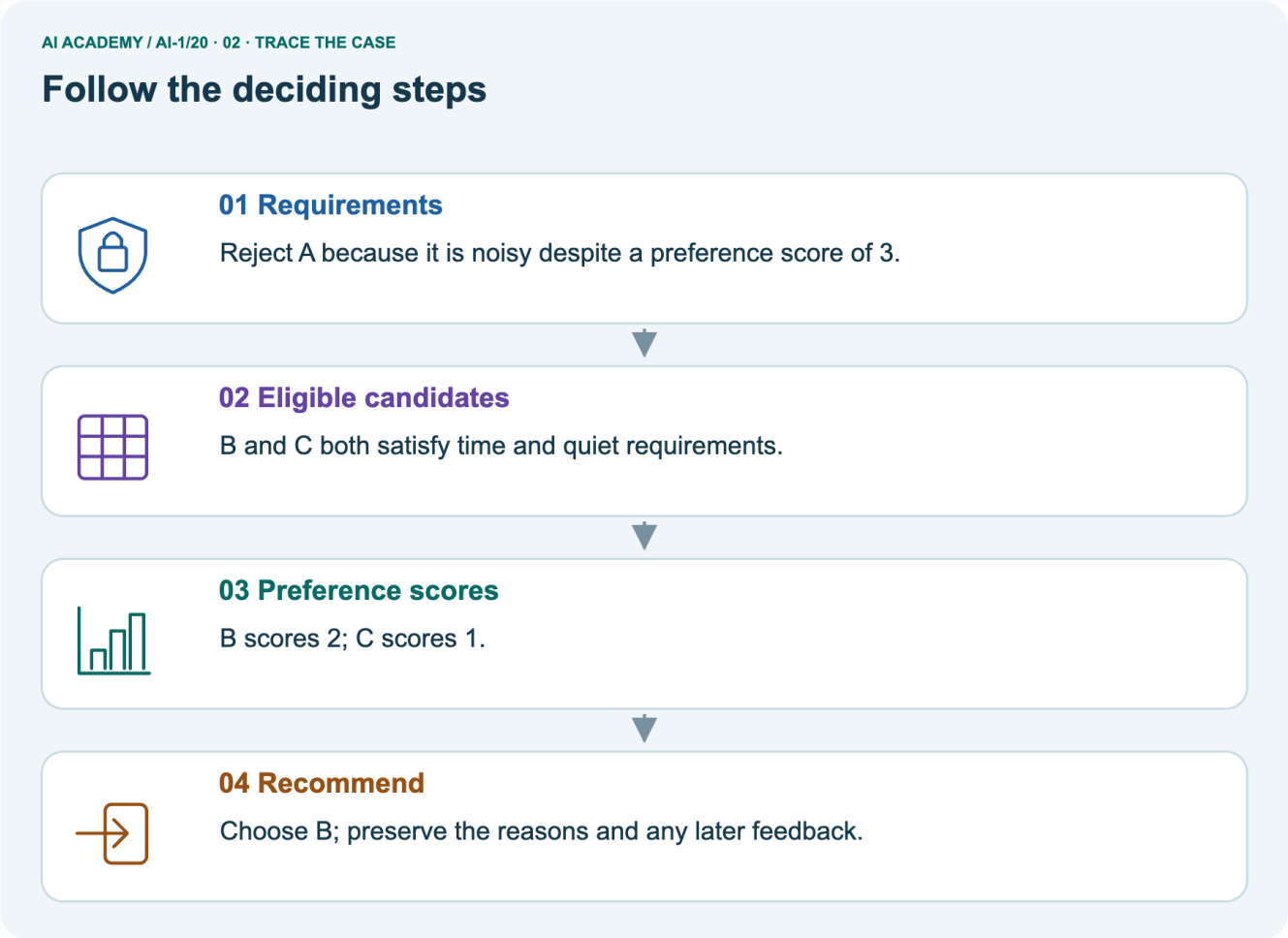
A fictional classroom break chooser recommends a suitable paper activity. Supplied facts: Must be quiet and take at most **12 minutes**. Score 2 for puzzles and 1 for solo.

A puzzle,solo,noisy,8 min; B puzzle,group,quiet,12 min; C drawing,solo,quiet,10 min.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



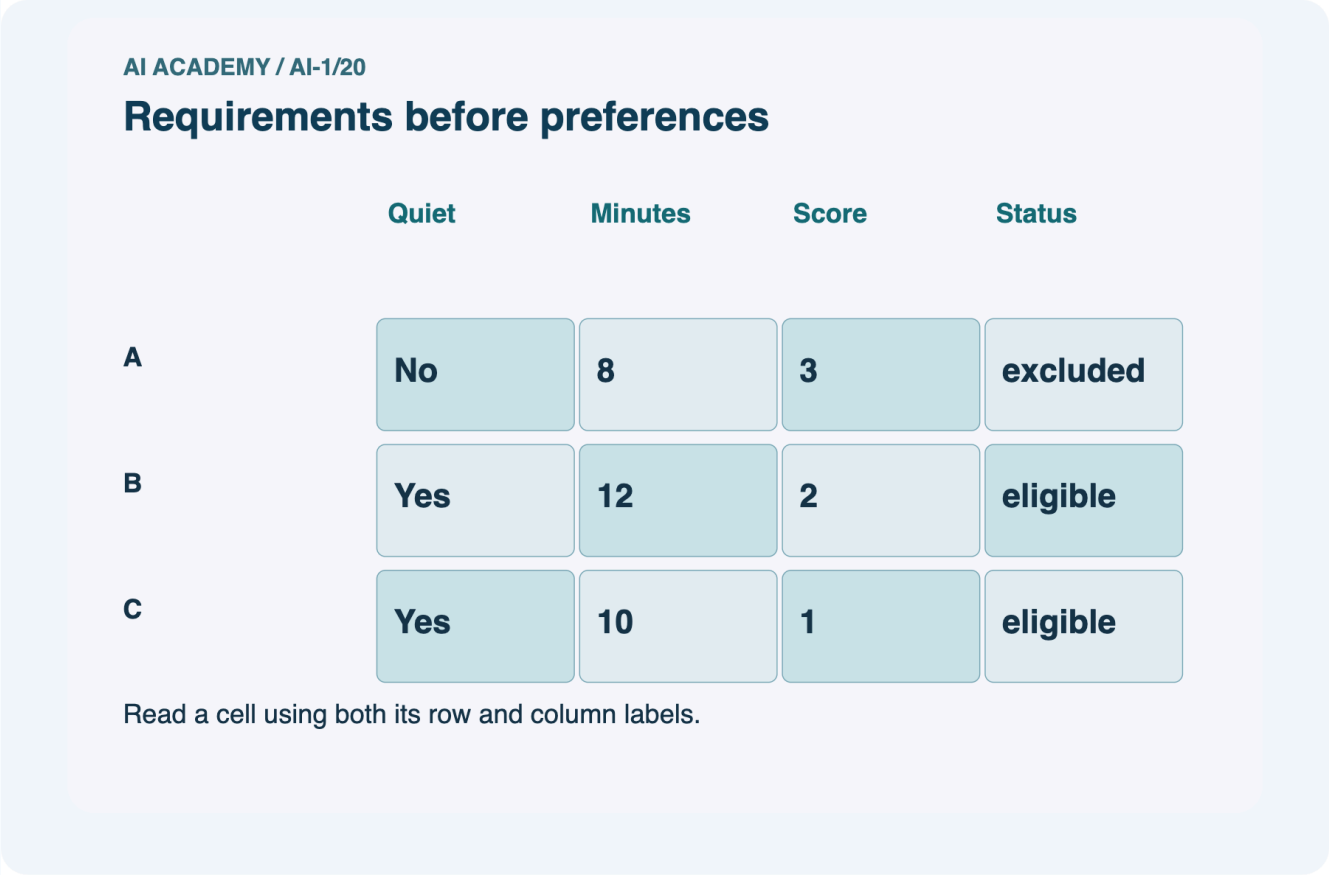
Follow the labelled picture with the matching explanation. Open the vector diagram

Requirements: Reject A because it is noisy despite a **preference** score of 3. → Eligible candidates: B and C both satisfy time and quiet requirements. → **Preference** scores: B scores 2; C scores 1. → Recommend: Choose B; preserve the reasons and any later feedback.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Requirements before preferences



Follow the labelled picture with the matching explanation. Open the vector diagram

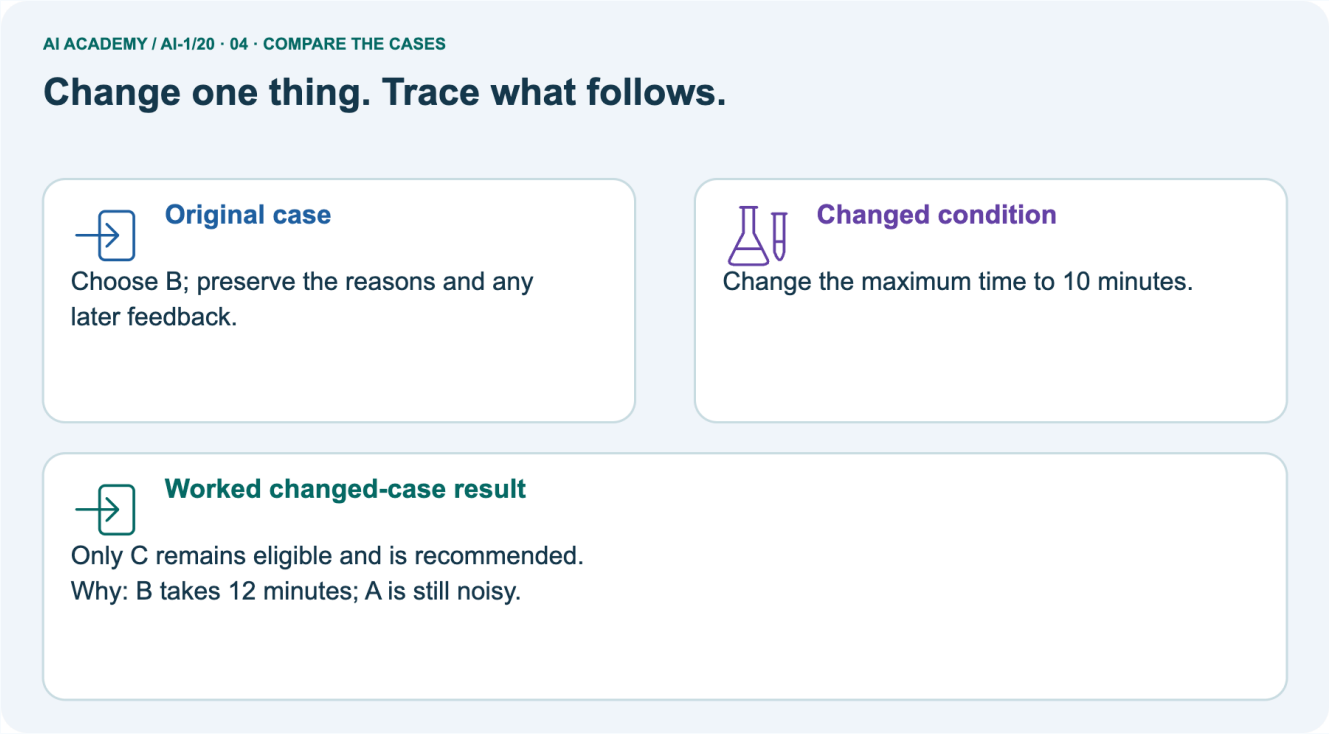
Requirements before **preference**s. A: Quiet = No, Minutes = 8, Score = 3, Status = excluded. B: Quiet = Yes, Minutes = 12, Score = 2, Status = eligible. C: Quiet = Yes, Minutes = 10, Score = 1, Status = eligible.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Choose B; preserve the reasons and any later feedback. Changed condition: Change the maximum time to 10 minutes. Expected result: Only C remains eligible and is recommended.

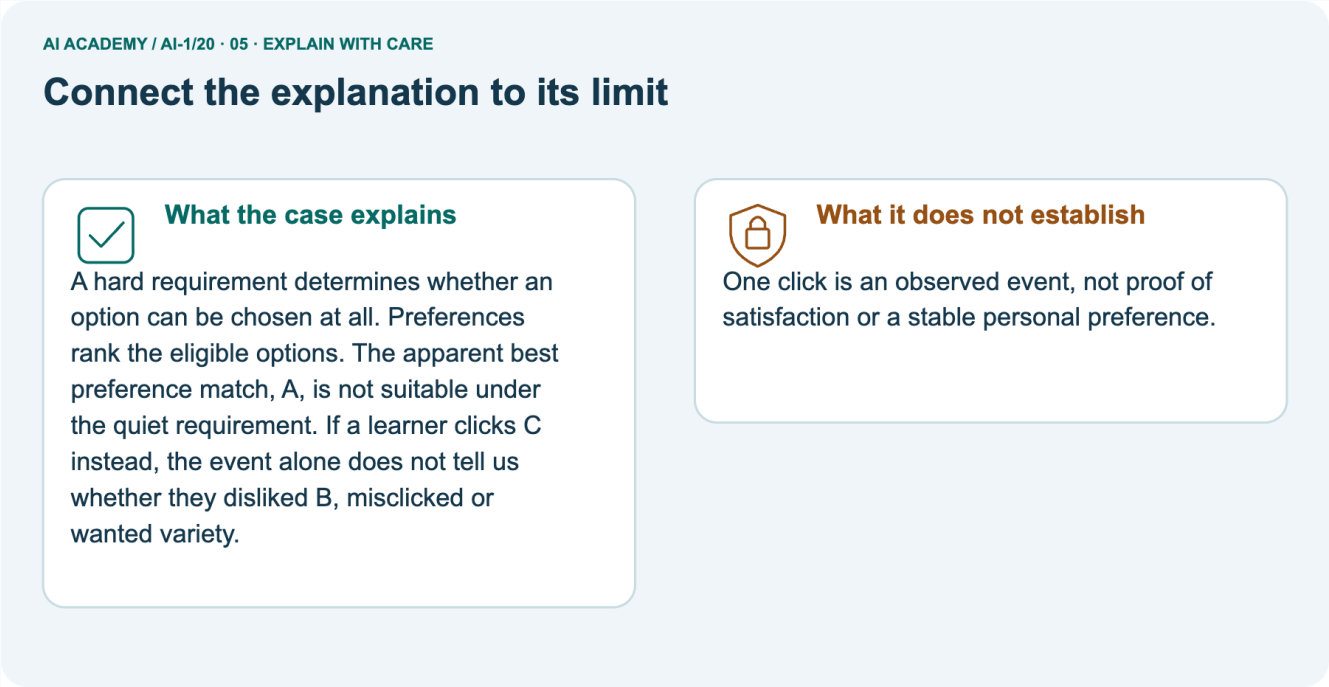
Reason: B takes **12 minutes**; A is still noisy.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: B takes **12 minutes**; A is still noisy.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A hard requirement determines whether an option can be chosen at all. **Preference**s rank the eligible options. The apparent best **preference** match, A, is not suitable under the quiet requirement.

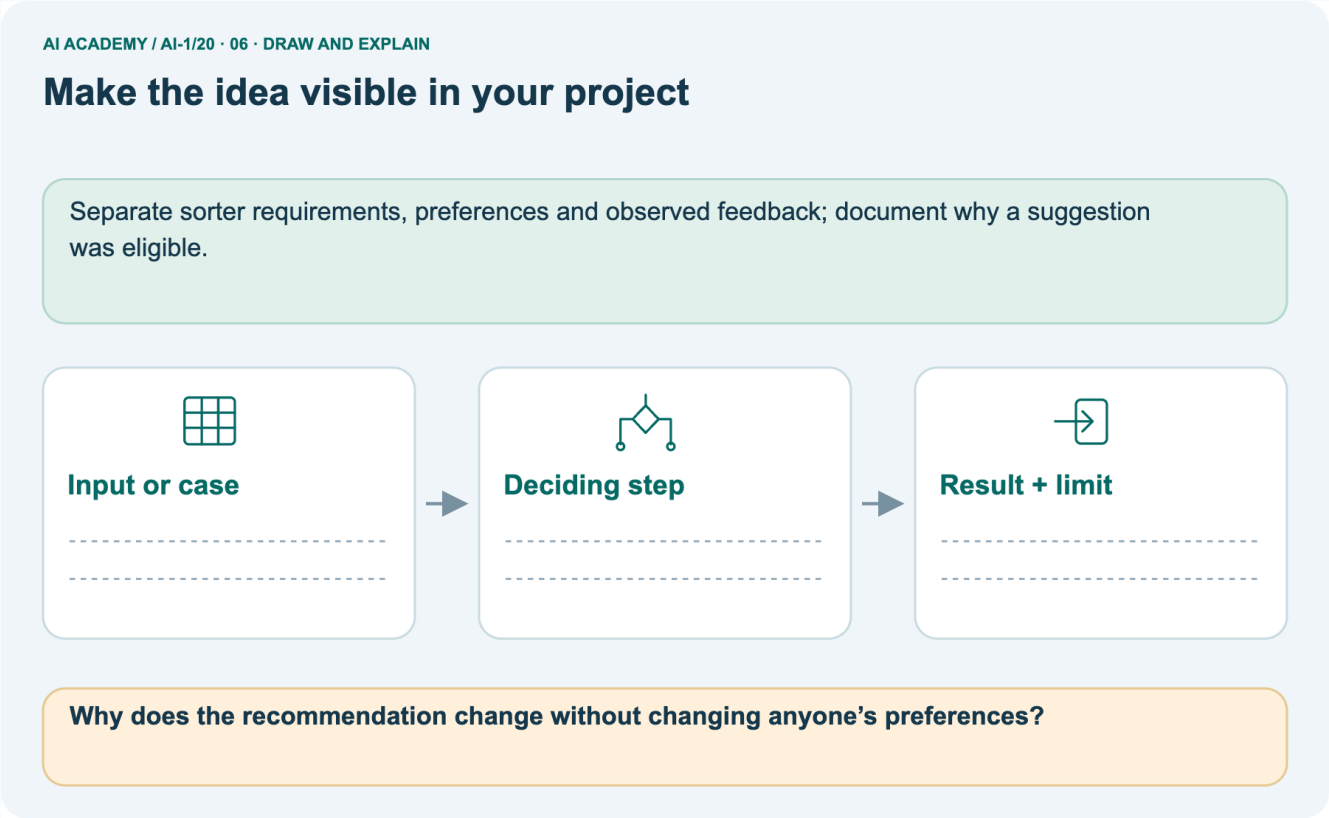
If a learner clicks C instead, the event alone does not tell us whether they disliked B, misclicked or wanted variety. Limit: One click is an observed event, not proof of satisfaction or a stable personal **preference**.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: One click is an observed event, not proof of satisfaction or a stable personal **preference**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Separate sorter requirements, **preference**s and observed feedback; document why a suggestion was eligible. Draw your **input**, deciding step, result and limit.

Supported question: Why does the **recommendation** change without changing anyone’s **preference**s?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The requirement changes **eligibility**. C becomes the only eligible choice even though B would still score more **preference** points.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Recommendations: the case

02. Follow the deciding steps

03. Requirements before preferences

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.